



Vel Tech
Rangarajan Dr. Sagunthala
R&D Institute of Science and Technology
DEEMED TO BE
University
(Estd. u/s 3 of UGC Act, 1956)
Avadi, Chennai

Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology
School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

Date: 26/03/2026

Real-Time Event Synchronization Engine

SDG:9– SDG Title :Industry, Innovation and Infrastructure

Presented By:

G.SAI MANIKANTA – VTU26808– Reg.No.23UECS0005

Courses Handling Faculty:

Dr. Hafizur Rahman–Assistant Professor

INTRODUCTION



1

- **Background of the problem**

People find it hard to track event updates

Missing important changes causes confusion

2

3

- **Why this project is important**

Makes event management simple and fast

Everyone gets updates at the same time

4

5

- **Real-world relevance**

Useful in colleges, offices and organizations

6

7

- **Brief overview of the system**

A website to create and manage events

Sends instant notifications on any change

Built using PHP, JavaScript and MySQL

8

9

PROBLEM STATEMENT



1

- Event updates are shared manually through messages

2

- People often miss important event changes

3

- No single platform to create, update and track events

4

- Difficult to notify everyone at the same time

5

- This leads to confusion, delays and miscommunication

6

- Tracking multiple events at once is very hard

7

- There is no automatic alert system for event status changes

8

- Organizations lose time and productivity due to poor event coordination

9

OBJECTIVE(S)



1

- To design and develop a real-time event management web application

2

- To implement instant notifications for event status changes

3

- To improve communication between event creators and participants

4

- To develop a secure login and registration system for users

5

- To implement a database to store and manage all event details

6

- To improve event tracking by providing filters and status updates

7

- To evaluate the performance of real-time polling every 4 seconds

8

- To develop an easy-to-use dashboard for managing all events

9

PROPOSED SYSTEM



1

- A web-based Real-Time Event Synchronization Engine (RESE)

2

- Users can create, update and track events from one platform

3

- Automatic notifications are sent when any event status changes

4

- Real-time polling updates the dashboard every 4 seconds

5

- Secure login system ensures only authorized users can access

6

- Filter events by status — upcoming, ongoing or completed

7

- Eliminates manual communication and reduces miscommunication

8

- Faster, smarter and more reliable than traditional event management

9

PROPOSED SYSTEM ARCHITECTURE



1

2

3

4

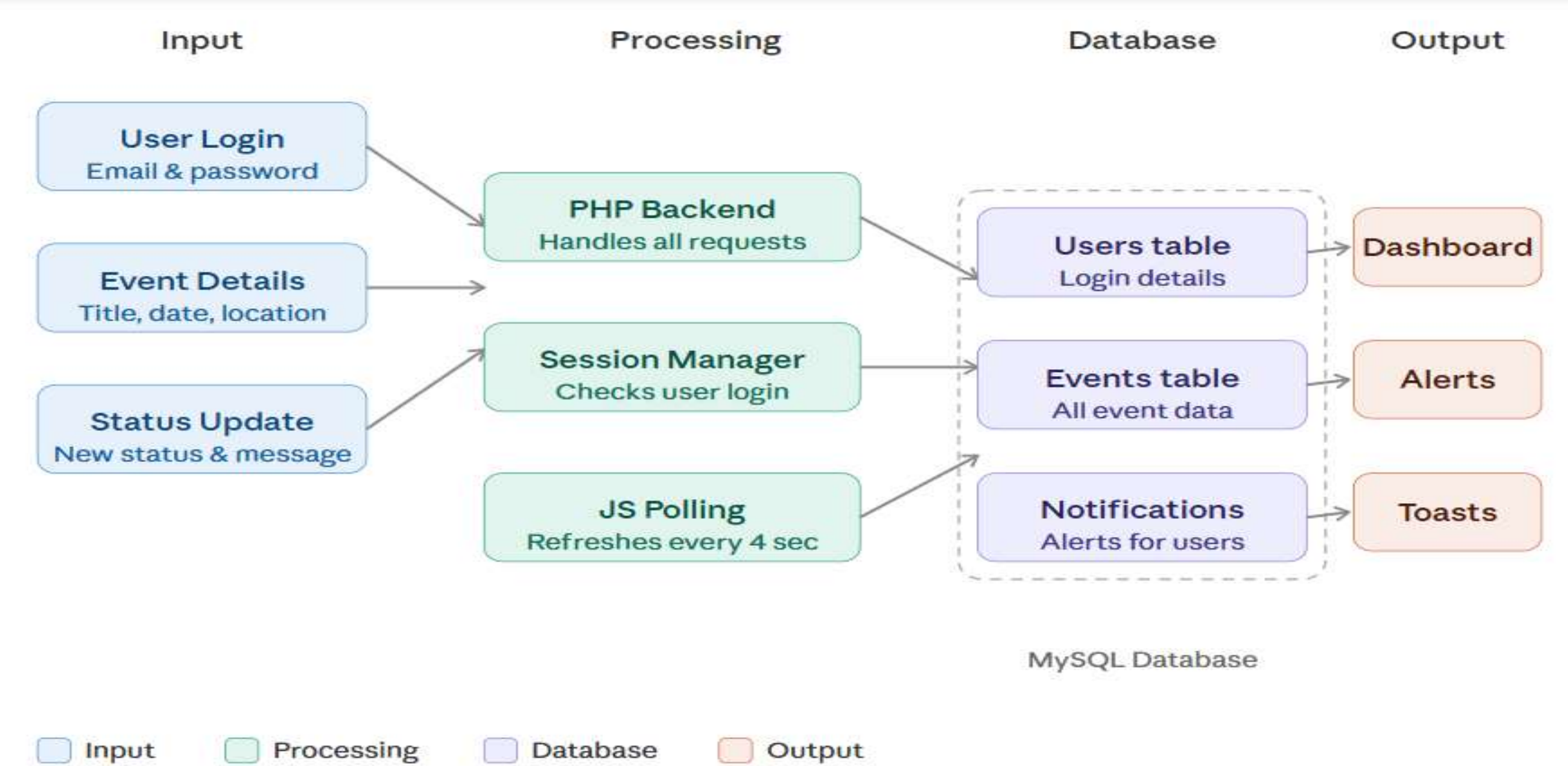
5

6

7

8

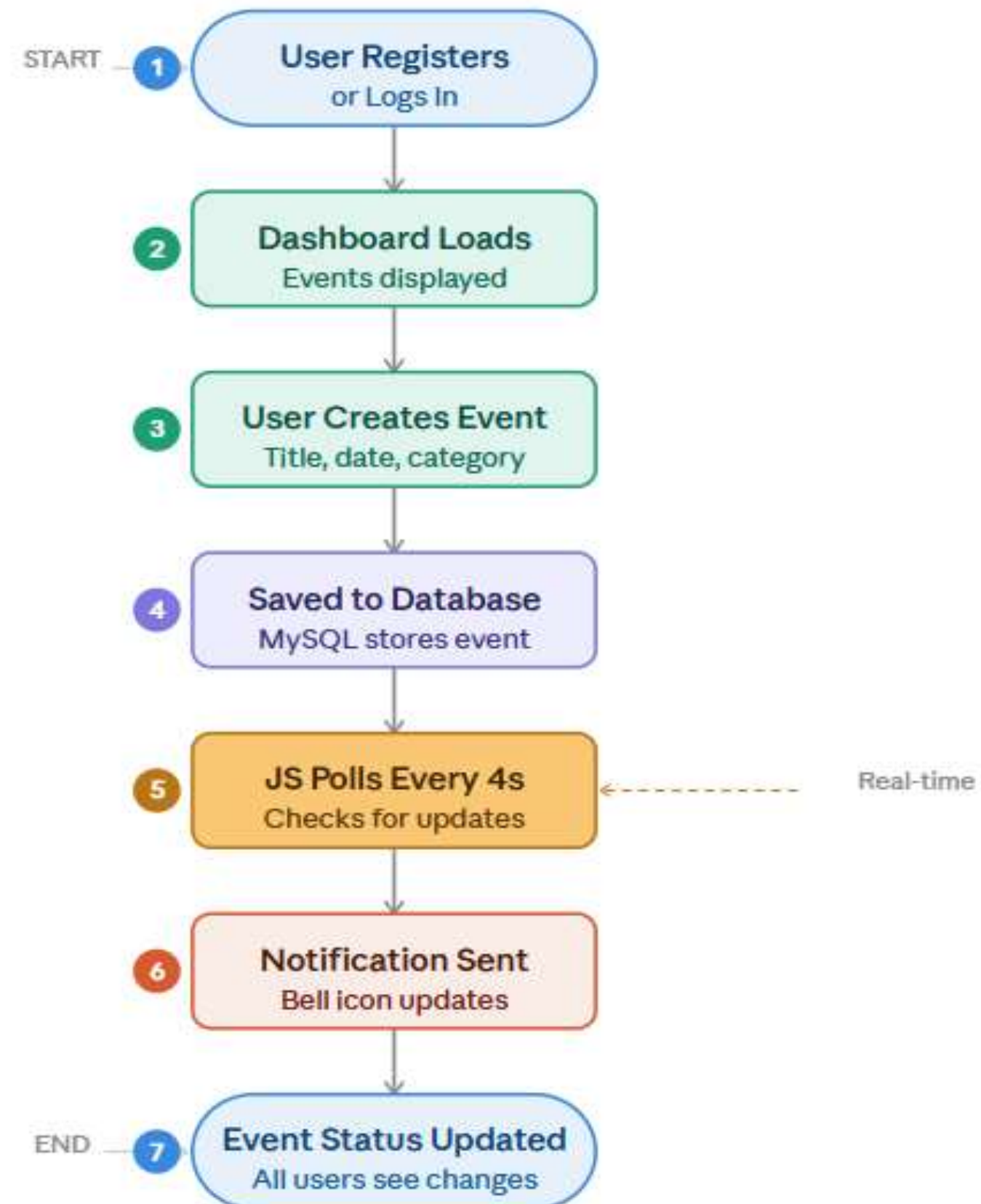
9



WORKFLOW



RESE Workflow



1

2

3

4

5

6

7

8

9

TECHNOLOGIES USED



- **Frontend:**

HTML5 — structure of web pages

CSS3 — styling and design

JavaScript — real-time polling and dynamic updates

- **Backend:**

PHP — handles all server-side logic

Session management — keeps users logged in securely

- **Database:**

MySQL — stores users, events, notifications and updates

phpMyAdmin — used to manage the database visually

- **Tools:**

VS Code — code editor

XAMPP — local server to run PHP and MySQL

Eclipse IDE — project management

- **Hardware/Software Requirements:**

Any computer with minimum 4GB RAM

Internet browser (Chrome, Firefox, Edge)

Windows OS

XAMPP v3.3.0

Any modern web browser

1

2

3

4

5

6

7

8

9

MODULE DESCRIPTION



1

Module 1: User Management

Handles user registration, login and logout with secure session-based authentication.

2

Module 2: Core Processing

Implements real-time event creation, status updates and JavaScript polling every 4 seconds to sync data automatically.

3

4

5

Module 3: Database Management

Manages data storage and retrieval using MySQL with 4 tables — users, events, event_updates and notifications.

6

7

8

Module 4: Reporting & Analytics

Displays live event dashboard with status filters, bell notification count and toast messages for better decision-making.

9

IMPLEMENTATION SCREENSHOTS



- <http://localhost/RESE> → Home page

1

2

3

4

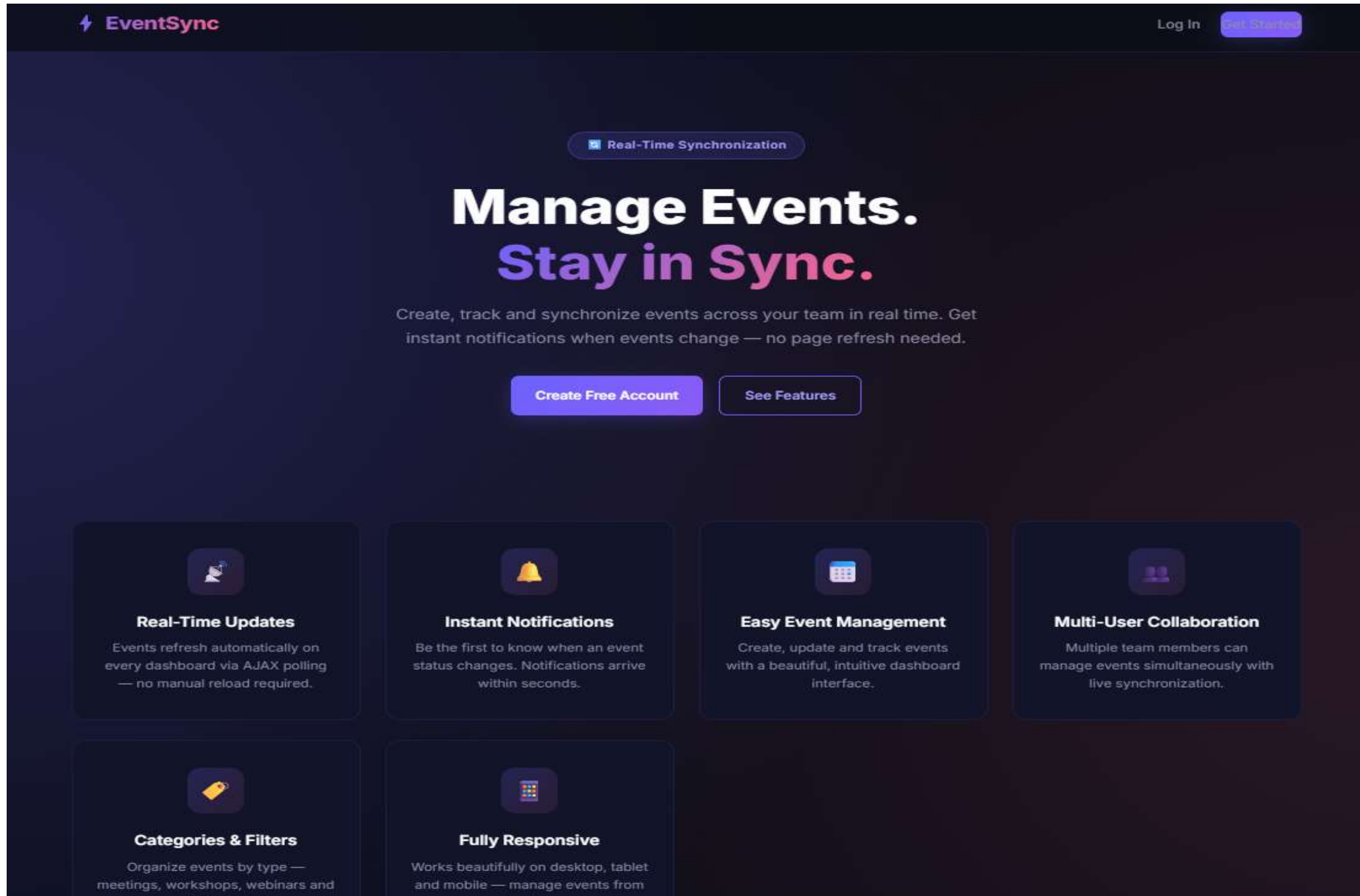
5

6

7

8

9



IMPLEMENTATION SCREENSHOTS



- <http://localhost/RESE/login.html> → Login page

1

2

3

4

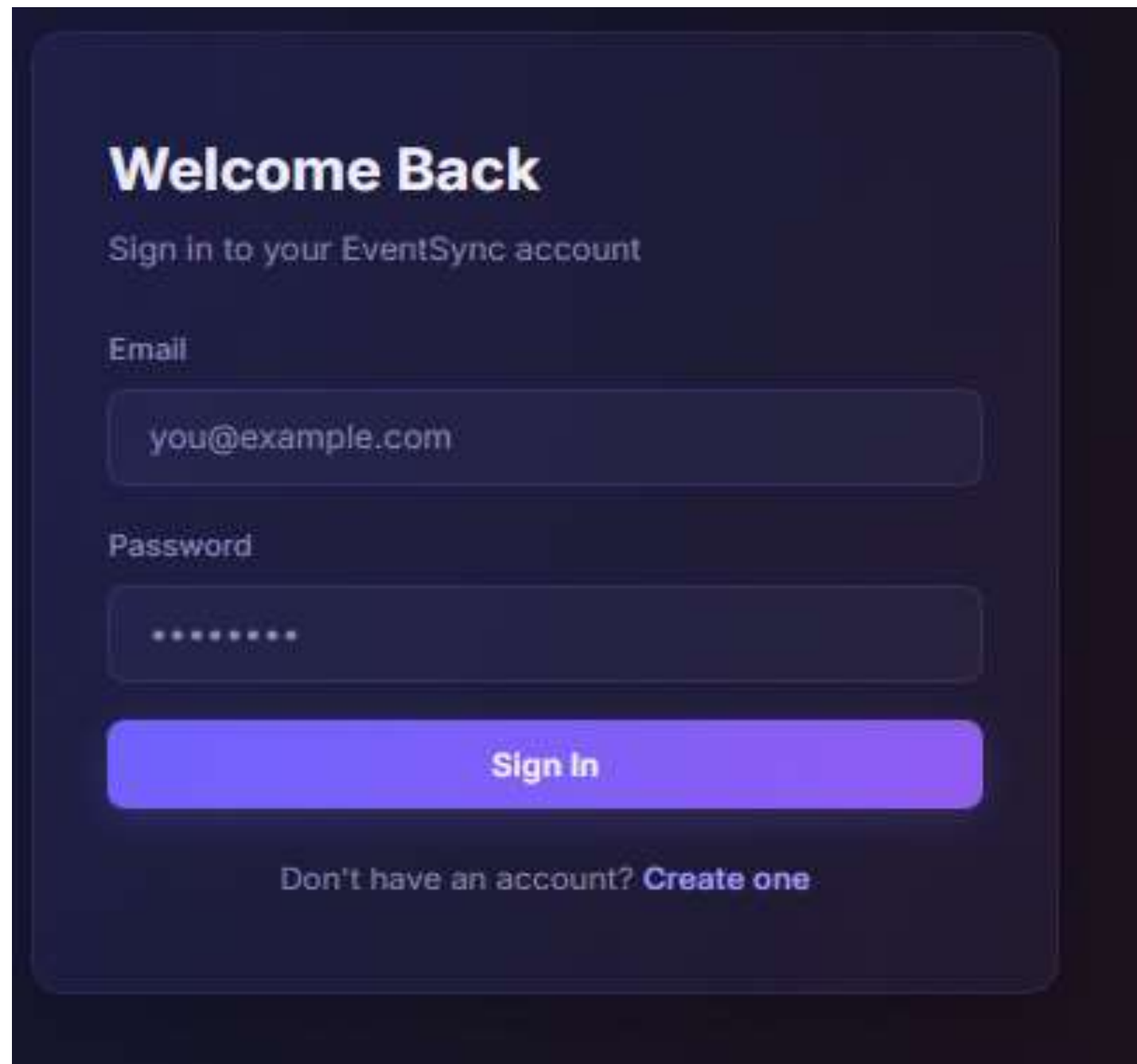
5

6

7

8

9

A screenshot of a login page for 'EventSync'. The page has a dark blue background. At the top, it says 'Welcome Back' in white, followed by 'Sign in to your EventSync account' in a lighter blue. Below this are two input fields: 'Email' with the placeholder 'you@example.com' and 'Password' with a masked password '.....'. A large blue 'Sign In' button is centered below the fields. At the bottom, it says 'Don't have an account? Create one' in white.

Welcome Back

Sign in to your EventSync account

Email

you@example.com

Password

.....

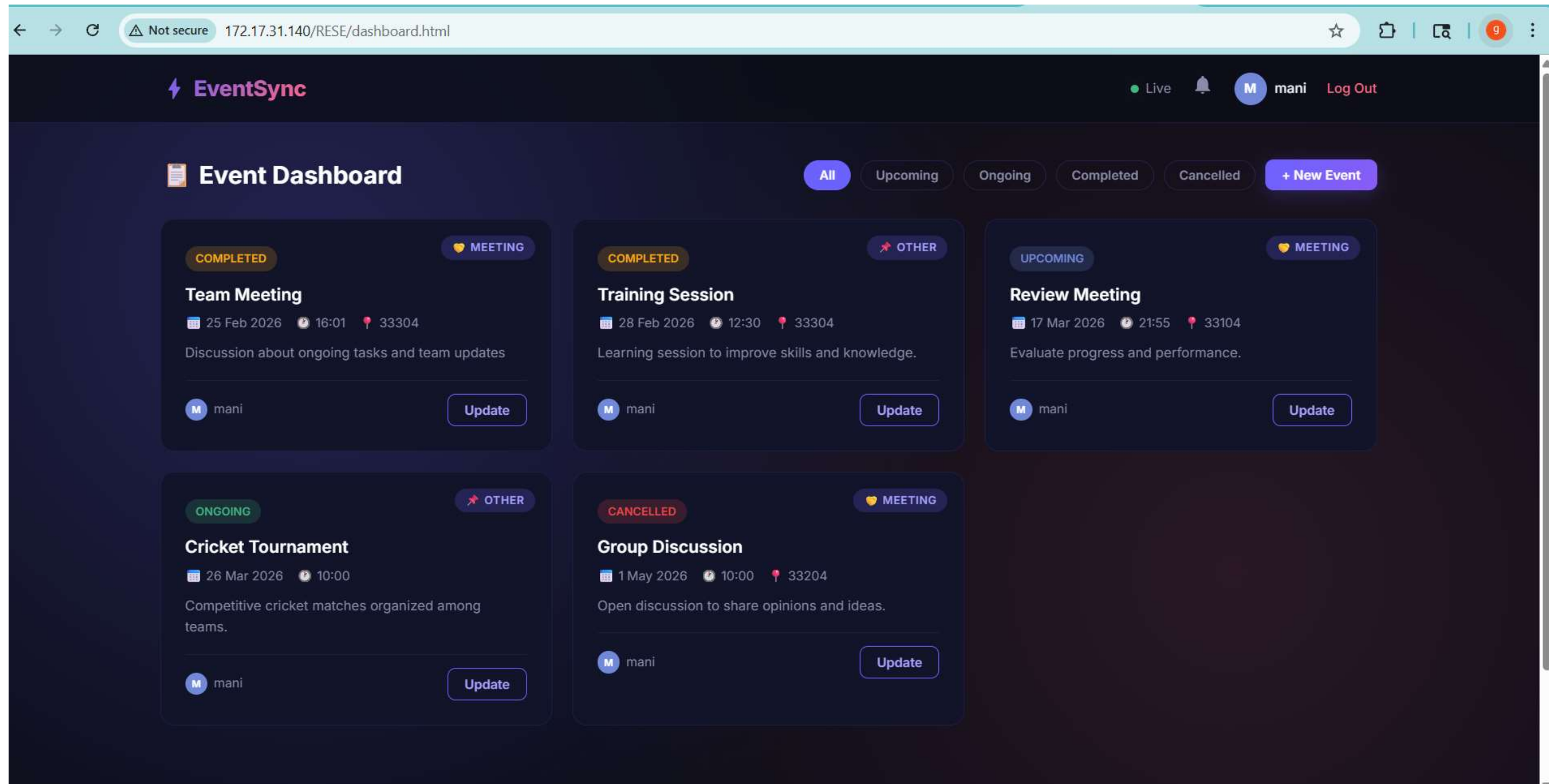
Sign In

Don't have an account? [Create one](#)

IMPLEMENTATION SCREENSHOTS



- <http://localhost/RESE/dashboard.html> → Dashboard



ADVANTAGES & LIMITATIONS



1

Advantages:

- Real-time updates without refreshing the page
- Easy to use for anyone — simple and clean interface
- Instant bell notifications when event status changes
- Secure login system protects user data
- All events stored safely in MySQL database
- Works on any device with a web browser

6

Limitations:

- Works only on local server (XAMPP) — not live on internet
- Requires XAMPP to be running every time
- No mobile app — only works on web browser
- Only one admin role — no different user permissions
- No email notifications — only in-app bell alerts
- Internet connection needed for browser to work

7

8

9



Thank you